

UK & EU Group Retail Rollup Process: Enhancement Proposal

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1 Executive Summary

The objective of this project is to develop a rollup process that can be transferred to an analyst with reasonable SQL and modelling experience. This should enable greater frequency of reporting — from annual to quarterly — and therefore improve the credibility and timeliness of results.

In order to achieve this, and based on learnings from the January 2026 rollup process, several foundational steps must be completed first.

The project is structured into three phases. The final phase remains open-ended, as there are systems and processes not yet fully understood; it is therefore expected that additional issues will surface requiring resolution in Phase 3.

A number of operational requirements detailed in Section 4 will be required. Ideally, these will be agreed rather than workarounds being done.

1.1 Project Phases

Phase 1: Current — March

Phase 1 is already underway and focuses on fact-finding and system setup. During this phase, work is ongoing with business units to review the data they currently produce, the frequency at which it can be delivered, and the cleansing steps required.

1. Map out data sources
2. Define acceptable data formats
3. Replicate the Guy Carpenter (GC) methodology to cleanse data
4. Agree analyst resource to assist with physically modelling the exposure data and to provide feedback on the data formats being produced

Phase 2: Mid-March — End of April

Replicate the January 2026 rollup post-modelling process into a cleaner, maintainable system. Bring together the outputs from Phase 1 into a coherent repository.

Phase 3: May — June

Resolve currently unknown issues or fill reasonable gaps uncovered during Phases 1 and 2. Additionally some of the data transacted between various systems at this point has gone through multiple ETL/ELT transformations. Phase 3 is a good place to take a step back and review them once more.

1.2 Project Timeline

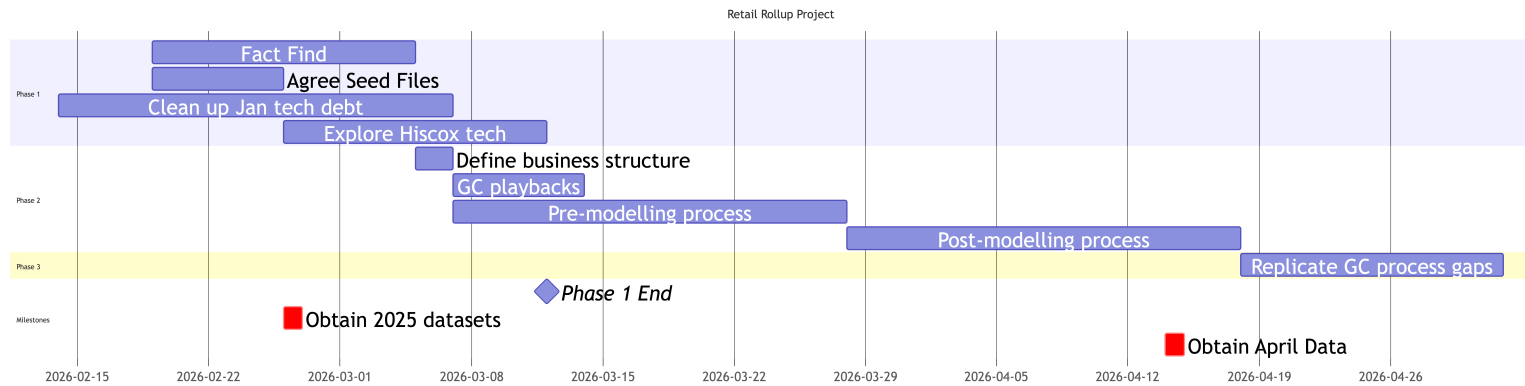


Figure 1: Project Timeline

2 Success criteria

1. Data sources have been mapped and documented ETL process to produce cat model inputs
2. Replication of GC work and playback reports
3. Cat model resource knows the defined data format and how to execute the modelling workflow
4. ELT process documented and defined to produce the required CDS inputs.

2.1 Nice-to-haves

Exposure based metrics such as Global Exposures, aggregates, various inputs for RDS, BSCR, PRA returns. It is unlikely a robust process to produce all of these can be accomplished. Instead the clean exposure datasets produced above would enable this analysis to be done in future.

3 Costs

Compute and storage costs are not expected to be significant. The rollup generated single-digit gigabytes of data and some of this was not optimised.

- Quarterly rollup size: 2–10 GB
- Runtime: 20 minutes – 2 hours (local execution)

Costs on Databricks for this will be small. Costs using a local database like Duckdb will be zero.

4 Governance Recommendations

The most important unresolved issue identified to date is the absence of Standard Operating Procedures (SOPs), runbooks, and checklists, combined with a lack of formal review processes. Without these controls, even a well-engineered system will be prone to failure.

The following actions are recommended for Hiscox management:

1. **Mandate the use of SOPs and runbooks**, stored and reviewed in a central version-controlled repository.
2. **Rigorously enforce managerial review practices** with multiple defined checkpoints throughout the process.

4.1 Background: Lessons from January 2026

The principal issues encountered during the January 2026 rollup stemmed from a lack of documentation and repeatable inputs:

- Inconsistent application of model settings and naming conventions year-on-year, which meant previous work could not be reproduced.
- Identifying where, when, and what had been done previously was extremely difficult in all cases.
- Files were scattered across network drives with no defined conventions and no recent documentation.
- Results were difficult to audit as Group Retail did not source the data directly, receiving it instead from Guy Carpenter.
- No defined review processes or checkpoints to prevent work proceeding on incorrect or unsatisfactory methodology and data.
- Friction and rework caused by legacy GUI-only tools; multiple issues were encountered, all of which are readily solvable.

Some of these issues are not caused by, nor solely fixable by, the Group Retail team. They do not own the underlying systems and are often several steps removed from the production of source data, which makes explaining results challenging.

4.2 Tertiary Outputs

There are a number of tertiary outputs produced as part of the rollup process:

- Internal RDS aggregates
- Lloyd's RDS
- BSCR / PRA exposure outputs

In almost all cases these were undocumented from previous years. Some calculations are suspected to require rebuilding — for example, the internal global exposures — as the input data may be corrupted and the calculation script itself is not maintainable.

4.3 Additional Outputs Not Currently Produced

Simple aggregation analysis at the state, city, and regional level is not currently produced. This makes analysis of model movements more difficult and prevents loss numbers from being contextualised.

Similar figures are frequently required to produce regulatory and tertiary outputs; as such, this represents a suite of outputs that Group Retail should aim to produce systematically.

5 Operational Requirements

In addition to the governance matters above, there are a number of relatively straightforward changes that Hiscox could implement to have an immediate positive impact on the smoothness of the Group rollup:

1. **Define responsibilities and timelines** for the delivery of raw data from business units.
2. **Clarify the requirements of the CDS system.** The lookup tables used in this system should be made immediately available to the retail team to minimise problems with importing invalid data to CDS.
3. **Chnges to CDS System** Any changes to the CDS system should follow their own governance process which should include informing the Group Retail team. Any changes to CDS can and will prevent smoothly uploading the information and precipitate workarounds which will have a high potential to end up being undocumented.
4. **Agree ownership of the modelling process** in Touchstone and Risklink. By extension, the data format to be delivered to, and produced by, the modeller will be formally agreed.
5. **Produce checklists and results templates** to aid managerial review and to give the analyst running the process clear criteria for completion.
6. **Implement modern analytical database tooling and version control.** These must become standard practice to ensure the systems do not deteriorate over time. Managers must enforce this or they will inevitably fall into disuse, ending back at undocumented and uncontrolled procedures.

6 Technical Architecture

This section details the technical components of the proposed process and the current approach.

6.1 Data Flow Overview

This section describes the current thinking around data flow based on what was done in January 2026 and meeting with the business units.

6.2 ETL Process

The disparate and changeable data sources, in their current state, may require an ETL process that cannot be fully contained within Databricks. This will be avoided where possible. Reduced flexibility is considered acceptable in this context — it enforces a clearly defined standard for what inputs are acceptable.

Note: The catastrophe modelling process is included within the ETL scope.

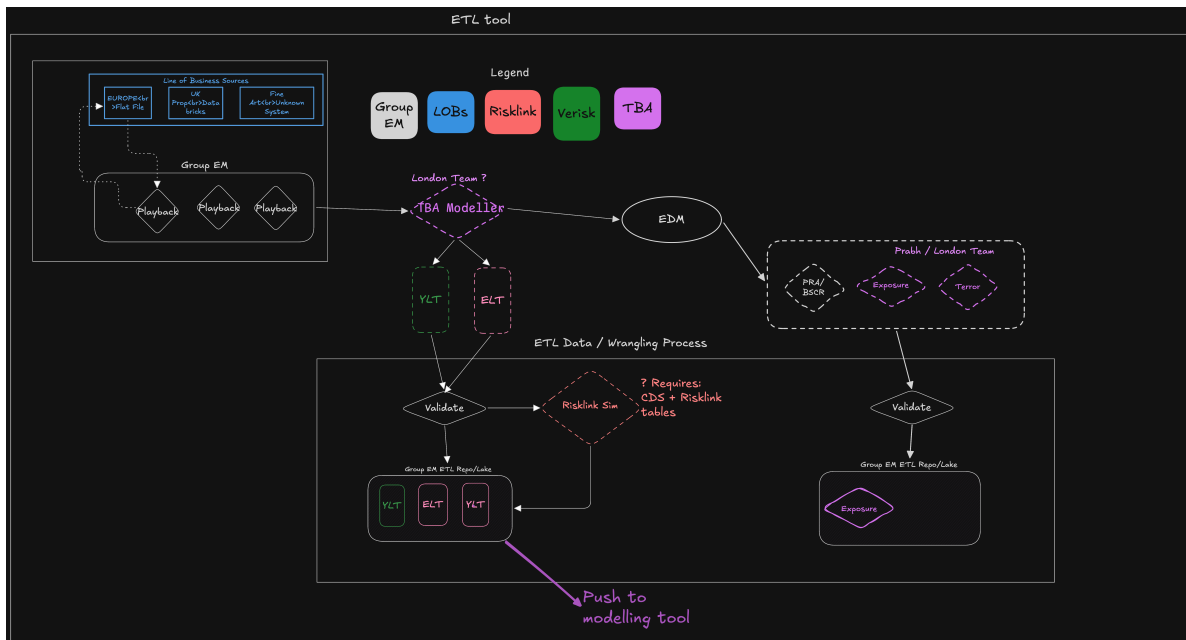


Figure 2: ETL Process

This schematic was produced prior to the adoption of Databricks; the technical implementation will change accordingly. The primary purpose here is to illustrate the workflow upstream of modelling and the application of the Hiscox View of Risk (VoR).

Key observations:

1. There are multiple playback points to business units, currently managed by GC.
2. There is no defined process for modelling risks using in-house systems.
3. An alternative simulation engine is in use to avoid reliance on Tomcat.
4. Pulling data from the back tables of CDS is required to correctly model Risklink and validate the inputs.

6.3 Pre-Modelling

A significant portion of the work involves generating exposure data, validating it, and playing it back to business units — a role currently performed by GC. Bringing this pre-modelling checking and data manipulation step in-house would expedite the rollup significantly.

Note: Hiscox Group is unlikely to see these intermediate outputs directly, but their accuracy is a prerequisite for reliable final results.

6.4 ELT and Data Modelling

Once the data sources have been loaded, the process transitions to a structured data modelling workflow. The goals of this stage are to:

- Systematically apply the Hiscox View of Risk (VoR), forecast factors, and various uplift factors
- Generate two output streams: one for the internal rollup and one for the dials-up view
- Optionally: generate aggregates and global exposures

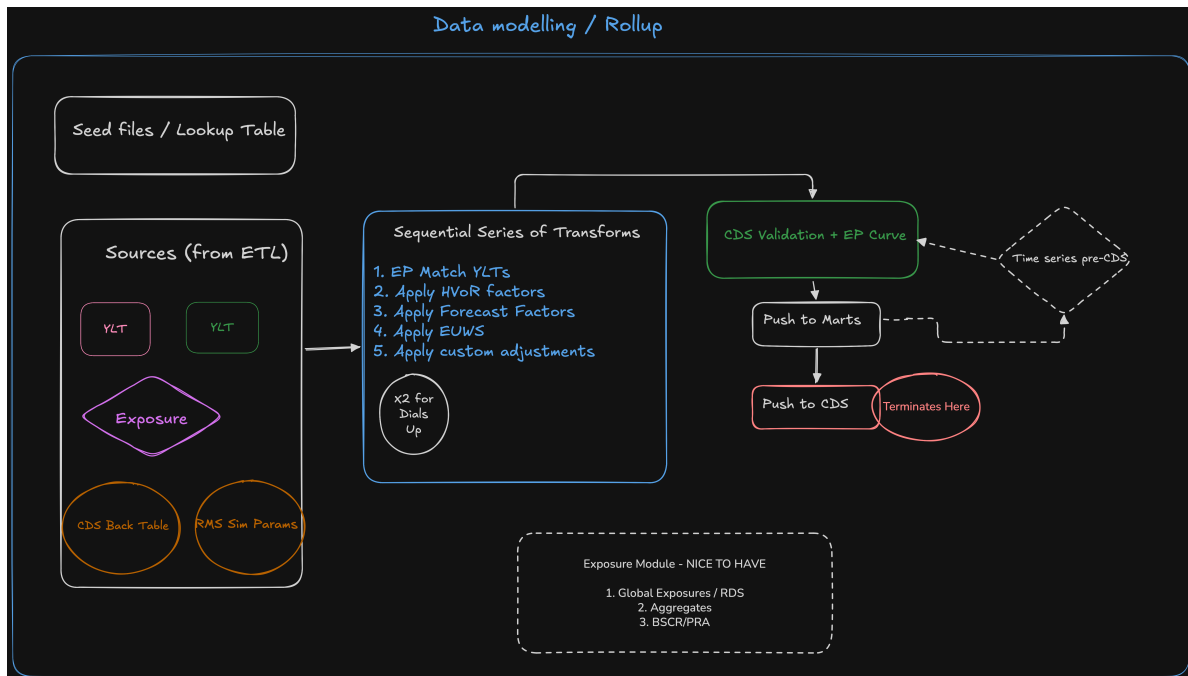


Figure 3: Post-Modelling ELT Workflow

7 Current Scope Limitations

Due to a small number of process constraints, it is not currently possible to deliver a fully end-to-end automated workflow. The specific limitations are as follows:

1. **Exposure data playback to business units.** This step arguably belongs with the business units themselves. The existing practice is for GC to load raw exposure data and return exposure reports with time series analysis.
2. **Production of model results via Risklink or Touchstone** currently requires manual file loading. Options for automation exist but will be constrained by the capabilities of the existing modelling platforms.
3. **The final upload to CDS is a manual process.** A live staging area, auto-loading pipeline, or web API integration is not currently available.

8 Review meetings

I suggest we hold fortnightly review meetings initially. As issues may crop up that require Hiscox to make decisions or assist in solving roadblocks.